LECTURE ACTIVITY 13: MURDEROUS NURSE?

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from "Introduction to Statistical Investigations," by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

1) Organize counts into a two-way table, when data are available on two categorical variables for the same set of observational units.
2) Calculate conditional proportion of successes, for different categories of the explanatory variable
3) Use conditional proportions to decide whether there is preliminary evidence of an association between the explanatory and response variables.
4) Create a segmented bar chart and mosaic plots to display data available on two categorical variables for the same set of observational units.
5) Calculate and interpret the difference in conditional proportions.
6) Calculate and interpret the relative risk.
7) Calculate and interpret the odds ratio.

Key Terms:

- Two-way table
- Conditional proportion
- Segmented bar chart
- Mosaic plot
- Relative risk
- Odds ratio

For several years in the 1990s, Kristen Gilbert worked as a nurse in the intensive care unit (ICU) of the Veterans Administration Hospital in Northampton, Massachusetts. Over the course of her time there, other nurses came to suspect that she was killing patients by injecting them with the heart stimulant epinephrine. Gilbert was eventually arrested and charged with these murders. Part of the evidence presented against Gilbert at her murder trial was a statistical analysis of 1,641 eight-hour shifts during the time Gilbert worked in the ICU. For each of these shifts, researchers recorded two variables: whether Gilbert worked on the shift and whether at least one patient died during the shift.

Question 1

Identify an observational unit in this study. (**Hint**: An observational unit is not a person. Think about what the sample size is.)

Question 2

Classify each variable as categorical or quantitative and determine which is the explanatory variable and which is the response variable.

(a) **Whether or not Gilbert worked on the shift:**

Select One: ☐ Categorical ☐ Quantitative
Select One: ☐ Explanatory ☐ Response

(b) **Whether or not at least one patient died during the shift:**

Select One: ☐ Categorical ☐ Quantitative
Select One: ☐ Explanatory ☐ Response

Question 3

Is this an observational study, or an experiment? Justify your answer.

So, what did the researchers find when they analyzed the 1,641 eight-hour shifts? They found that Gilbert worked during 257 of the shifts, and she did not work during the remaining 1384. They also found that at least one patient died during 74 of the shifts, and no death occurred during the other 1,567 shifts. Among the 74 shifts during which at least one patient died, 40 were shifts on which Gilbert worked, and 34 were shifts on which Gilbert did not work.

Concept Overview

When analyzing data on two categorical variables, a useful first step is to organize the data into a **two-way table** of counts. In this case both variables have only 2 categories, so we produce a 2×2 table.

Question 4

Fill in the missing counts in the following two-way table. **Hint:** Start by entering 40 and 34 into the appropriate cells. Then make sure that both rows and both columns add up to the indicated totals.

	Gilbert working on shift	Gilbert not working on shift	Total
At Least One Patient Died During the Shift			74
No Patients Died During the Shift			1567
Total	257	1384	1641

Question 5

Suggest a better way to compare the outcomes of the shifts on which Gilbert worked and the shifts on which she did not work than simply comparing 40 to 34. Explain why your answer is more informative.

Concept Overview

The next step in analyzing categorical data in a two-way table is to calculate **conditional proportions**. A conditional proportion simply means that you consider the counts in only one category of the explanatory variable at a time.

Question 6

(a) Calculate the conditional proportion of Gilbert shifts on which at least one patient died. (**Hint**: Put the total number of Gilbert shifts in the denominator of your fraction.)

(b) Calculate the conditional proportion of non-Gilbert shifts on which at least one patient died.

Visual Explorations

In many cases when comparing two variables, a good set of visualizations is enough to be able to start to see patterns. That is, even without sophisticated statistical tools, we can start to examine whether or not there is an association between whether or not Gilbert was on shift and whether or not at least one patient died.

Let's explore a few of these visualizations now.

Segmented Bar Graph

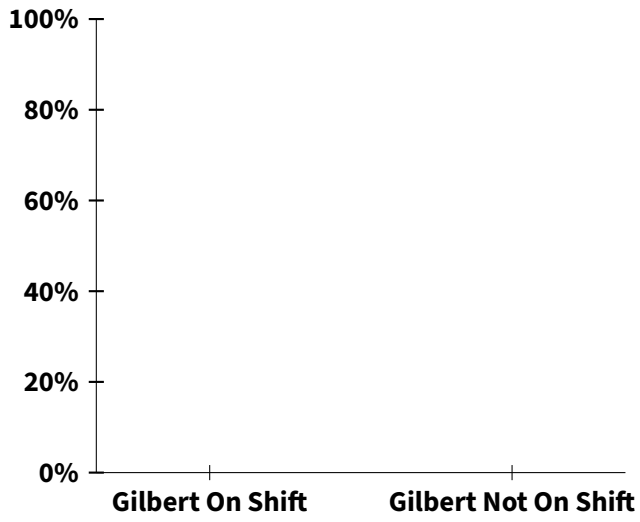
Concept Overview

A **segmented bar graph** is an appropriate display for graphing data in a two-way table. Such a graph contains rectangles of total height 100% for each category of the explanatory variable. Then segments divide up each rectangle according to the conditional proportions of the response variable categories. With this type of graph, you can display the association between the two categorical variables by showing the conditional proportions of success and failure across the explanatory variable groups.

- If the variables have absolutely no association, then the breakdown of segments is identical for across all of the bars.
- If the variables have absolutely no association, the conditional proportions are identical for all explanatory variable groups.

Question 7

Fill in the following segmented bar graph to display the data from this study. First draw two rectangles, one for Gilbert shifts and one for non-Gilbert shifts, each with total height 100%. Then draw a line in each rectangle corresponding to the conditional proportion of shifts in which a patient died, as you calculated in Question (4) above.



Mosaic Plots

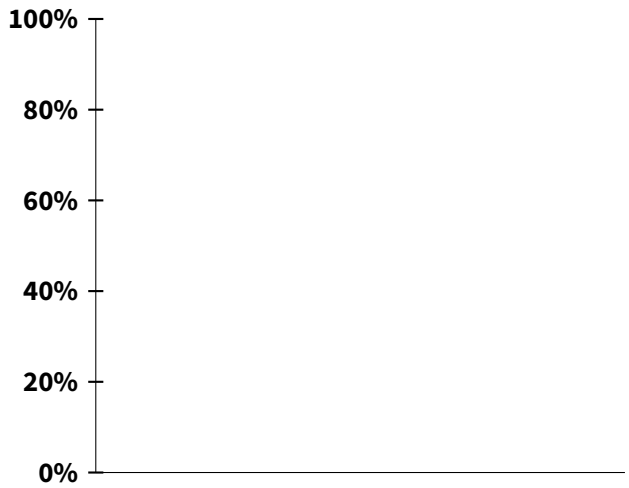
Concept Overview

Another visual is a **mosaic plot**, which allows the widths of the bars to represent the size of each group in the sample. A mosaic plot also makes the widths of the rectangles proportional to the sizes of the categories.

Question 8

Fill in a mosaic plot to display the data from this study:

- Start by drawing two rectangles, one for Gilbert shifts and one for non-Gilbert shifts, but make the rectangles have different widths.
  - Specifically, make whichever category (Gilbert on Shift, or Gilbert Not on Shift) have a smaller width.
- Then draw a line in each rectangle corresponding to the conditional proportion of shifts in which a patient died, as you calculated in Question (4) above.



Question 9

Do the conditional proportions, the segmented bar graph, and the mosaic plots appear to provide evidence that at least one patient dying was more likely to happen on a Gilbert shift than on a non-Gilbert shift? In other words, does there appear to be a tendency for Gilbert shifts to have at least one death more often than non-Gilbert shifts? Explain.

Question 10

Without doing any further analysis, do you consider the difference between the conditional proportions to be striking, worth reporting to a jury?

Numerical Explorations

A next step in this analysis is to produce a single number (statistic) to summarize the data. We will cover 3 different statistics in this part of the activity: difference in sample proportions, relative risk, and odds-ratio.

Statistic 1: Difference in Proportions

So far in this activity we have obtained two summary measures:

- The conditional proportion of Gilbert shifts on which at least one patient died: 40/257 = 0.156
- The conditional proportion of non-Gilbert shifts on which at least one patient died: 534/1,384 = 0.025

If we wanted to change these 2 numbers into a single statistic, we could just compute the **difference** in these proportions.

Question 11

Calculate the difference in conditional proportions of death between Gilbert and non-Gilbert shifts. Does the value of this difference strike you as noteworthy?

Question 12

Write a sentence interpreting the value you obtained in Question (11) above.

Statistic 2: Relative Risk

Another way that we can summarize the two proportions into a single statistic is to divide one proportion by the other.

Concept Overview

Relative risk is the ratio of two conditional proportions. It indicates how many times greater the risk of an outcome is for one group compared to another.

Question 13

Now calculate the relative risk. **Tip:** put the larger value in the numerator.

Question 14

Write a sentence interpreting the value you obtained in Question (13) above.

One drawback to using relative risk is that it can convey different impressions of how strong the relationship is depending on how the calculation is set up. That is, the value of the relative risk depends on which variable is explanatory, which is response, and which category is called "success." This can be confusing.

Statistic 3: Odds Ratio

In certain observational studies, samples of subjects are taken based on their status according to the *response* variable. Such studies are called **case-control** studies. Because the sample sizes in a case control study are dependent on the response variable, examining differences in proportions and relative risk ratios doesn't make a whole lot of sense (at least, in terms of trying to get a real-world risk associated with Gilbert being on shift). Instead, we use a new type of statistic: the odds ratio.

Let's build up to the notion of an odds ratio. First recall that the **proportion** of successes (in a group) is a ratio between the number of successes and the total number of observations (in the group):

$$\text{proportion of success} = \frac{(\text{number of successes})}{(\text{number of successes}) + (\text{number of failures})} = \hat{p}$$

The **odds** of success is a ratio between the number of successes *and the number of failures*:

$$\text{odds of success} = \frac{(\text{number of successes})}{\text{number of failures}} = \frac{\hat{p}}{1-\hat{p}}$$

Concept Overview

The **odds ratio** is the ratio of odds:

$$\text{odds ratio} = \frac{\text{odds for group 1}}{\text{odds for group 2}}$$

- The odds ratio indicates how many times greater the odds of success are for one group compared to another
- A major advantage of the odds ratio is that any 2×2 table has only one possible value of the odds ratio. That is, the odds ratio does not depend on the explanatory/response distinction, or the success/failure distinction (assuming you follow the convention that the calculation is set up so the odds ratio is at least 1).
- The disadvantage of the odds ratio is that it's harder to interpret ("higher odds" instead of the more natural "more likely").

Question 15

Calculate and interpret the odds ratio comparing the odds of death for the Gilbert shifts to the odds of death for the non-Gilbert shifts. Interpret this value.

Question 16

Calculate and interpret the odds ratio comparing the odds of staying alive for the Gilbert shifts to the odds of staying alive for the non-Gilbert shifts. Interpret this value.

Question 17

Even if the difference between these conditional proportions of death is found to be highly unlikely to have happened by chance, is it reasonable to conclude that Kristen Gilbert was responsible for (i.e., caused) the increased chance of death on her shifts? Explain your answer. (**Hint**: Remember your answer to Question (3) about what kind of study this is.)

Question 18

Suggest an alternative explanation that the defense attorneys might use to explain the large difference in death proportions between the two groups.

Question 19

(**Optional**) Was it enough? That is, navigate to Google (or your favorite search engine) and see what the "final" outcome of the trial was.